Governance and Accountability by Design in Agent-Based AI

Guardian Health

Comprehensive Case Study: Enterprise Architecture · Data Architecture · Agentic AI Architecture

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1. Introduction

1.1 What We Are Building

GUARDIAN-Health is an agentic AI system for the continuous monitoring of patient safety across a multi-centre hospital network. The system detects, classifies, and manages clinical adverse events and pharmacovigilance signals, coordinating responses across clinical departments, pharmacy, medical leadership, and regulatory authorities.

The architectural purpose of this case study extends well beyond its functional scope. It is an executable demonstration that **governance, traceability, and accountability** in agentic AI systems cannot be retrofitted. They must be woven into the design from the first line of code, the first data decision, and the first node of the graph.

The system implements three interconnected agentic modules:

- **Module 1: VIGIL (Vigilance and Detection).** Agents that continuously monitor clinical signals, prescriptions, laboratory results, and nursing records in real time, identifying anomalous patterns that may indicate adverse events.

- **Module 2: ASSESS (Evaluation and Classification).** A crew of agents that evaluates the severity, causality, and reportability of each detected signal, producing structured recommendations with full reasoning traceability.

- **Module 3: RESPOND (Response Coordination).** Coordination agents that manage communication between clinical actors, escalate according to defined HITL protocols, and generate the required regulatory documentation, including notifications to the relevant National Competent Authority (NCA) and safety reports in E2B R3 format.

1.2 Why This Use Case

Pharmacovigilance and patient safety in hospital environments represent one of the most demanding domains for autonomous agentic AI systems. This choice is deliberate and architecturally motivated.

- **Critical consequences of decisions.** An agent that misclassifies a serious adverse event as minor may delay life-saving interventions. This makes governance of agentic decisions not optional, but a hard design constraint.

- **Regulatory density.** The EU General Data Protection Regulation (GDPR), the EU AI Act, the European pharmacovigilance regulations (Regulation (EU) No 1235/2010 and its

amendments), and the guidelines of the European Medicines Agency (EMA) create an environment in which every agent action must be justifiable, traceable, and auditable.

- **Data heterogeneity.** Clinical data combines structured records (HL7 FHIR, prescribing systems), semi-structured content (clinical notes, discharge summaries), and unstructured text (nursing narratives, internal communications). This requires coordinated use of relational, graph, and vector databases.

- **Multiple stakeholders with divergent interests.** Clinicians, hospital administration, regulatory authorities, patients, and insurers each hold different perspectives and levels of trust regarding system decisions. This demands a governance design capable of satisfying multiple accountability frameworks simultaneously.

- **Demonstrable real-world value.** Active AI-driven pharmacovigilance is identified as a strategic priority by both the EMA and the World Health Organisation (WHO). A case study in this domain carries direct relevance to real projects in the healthcare sector across Europe and beyond.

1.3 Technologies, Frameworks, and Architectural Rationale

Technology selection is not arbitrary. Each component addresses a specific architectural requirement for governance, traceability, or agentic capability.

Agentic Orchestration Stack

- **LangChain** serves as the foundational abstraction layer for LLM integration, tools, and processing chains. It provides connectors to Groq, vector store abstractions, and a structured prompt system with native traceability support. Its selection is justified by its production maturity and the breadth of its integration ecosystem, which is critical when the system must connect with multiple heterogeneous clinical data sources.

- **LangGraph** is the stateful agent orchestrator. In a pharmacovigilance system, state is central: an adverse event passes through multiple states (detected, under evaluation, classified, reported, closed) with transitions that must be explicit, auditable, and reversible under human supervision. LangGraph implements this flow as a directed graph with persistent state management, native HITL support via node interrupts, and checkpointing capability that allows the flow to be rewound for audit purposes.

- **CrewAI** manages the evaluation crews in the ASSESS Module. When multiple specialised agents (pharmacologist, clinician, regulatory expert) must collaborate on signal assessment, CrewAI provides role abstractions, structured delegation, and a consensus process. Its model of tasks with Pydantic-typed outputs ensures that each step of collective reasoning produces structured, traceable artefacts.

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- **AutoGen** is employed for adversarial evaluation and protocol simulation. AutoGen's capacity to create multi-agent conversations with formally defined user and assistant roles is particularly useful for simulating a debate between a classification-proposing agent and a critical reviewing agent, replicating the peer review process in safety assessment.

Data Infrastructure

- **Neo4j** is the graph database for the clinical relationship map: patients, medicines, prescribers, diagnoses, adverse events, and treatment protocols. Pharmacovigilance signals are inherently relational: detecting that a medicine causes an adverse reaction in patients with a specific combination of comorbidities requires traversing relationship graphs, not querying flat tables. Pattern detection algorithms in Neo4j (via Cypher and APOC) allow identification of signals that would be invisible in flat relational models.

- **PostgreSQL** is the relational database for transactional data, audit logs, agentic flow state (via the LangGraph checkpointer), and master data. Its selection over NoSQL alternatives is justified by its ACID transaction support, which is critical when recording regulatory decisions with legal implications.

- **Pinecone** provides the cloud vector store for semantic searches over medical literature, clinical guidelines, drug data sheets, and hospital protocols. Unlike Qdrant (also used locally for development), Pinecone in production offers managed scalability and SLAs appropriate for clinical environments.

- **Qdrant** is used in the local development environment as an alternative to Pinecone, maintaining the same LangChain abstraction interface to guarantee portability between environments.

Integration and API Layer

- **FastAPI** exposes system services as REST APIs with automatic OpenAPI documentation. In a hospital environment, interoperability is non-negotiable: Electronic Health Record (EHR) systems, prescribing systems, and regulatory reporting portals must be able to integrate with GUARDIAN-Health through well-defined, versioned API contracts. FastAPI with Pydantic guarantees strict type validation at system boundaries.

- **Solace Agent Mesh** implements event-driven messaging for asynchronous coordination between modules. In a real hospital system, events (a new prescription, an abnormal laboratory result, a nursing alert) arrive asynchronously from multiple source systems. Solace provides an event bus with delivery guarantees, publish-subscribe pattern support, and the capability to integrate the agentic system with the existing hospital messaging ecosystem (HL7 v2, FHIR R4).

- **n8n** (self-hosted, free tier) acts as a workflow engine for non-agentic automation: scheduled notifications, periodic report generation, reference data synchronisation, and triggers for routine

regulatory processes. Its visual interface allows non-technical pharmacovigilance team members to configure and monitor workflows without modifying code.

Inference and Observability

- **Groq** provides ultra-low-latency LLM inference with `llama-3.3-70b-versatile` as the primary model. In a clinical context, latency matters: a safety alert system that takes minutes to process a signal is of no use in an emergency setting. Groq allows complex signals to be processed in seconds using the reasoning capability of 70B-parameter models.

- **LangSmith** is the observability layer for all agentic traces. Every LLM invocation, every tool execution, every state transition, and every HITL decision is recorded in LangSmith with its complete context. In a regulated system, this is not a debugging tool; it is a governance component. The LangSmith trace is part of the audit trail.

- **HuggingFace** (`nomic-ai/nomic-embed-text-v1`, 768 dimensions) provides the embeddings for semantic indexing of clinical documents. The choice of an open-source embedding model guarantees reproducibility and auditability of the vectorisation process.

1.4 The Central Principle: Governance from Day Zero

This case study exists to demonstrate a principle that the industry learns, invariably, the hard way: **in agentic AI systems, retroactive governance does not exist**.

An agentic system that begins without governance develops, within weeks, a governance debt that cannot be repaid:

- Agent decisions become opaque because no one designed the audit trail structure from the outset.
- Data used for context has no lineage because metadata was never established at the source.
- Responsibilities are ambiguous because human oversight roles were not defined as part of the agentic flow.
- Regulatory compliance becomes a forensic nightmare because records lack the structure that auditors require.

In the chapters that follow, every architectural design decision is motivated by a governance requirement. Governance is not a layer added at the end. It is the backbone of the system.

2. Case Study: GUARDIAN-Health

2.1 Context and Problem Statement

HealthBridge Network is a fictional yet architecturally realistic hospital consortium comprising three tertiary-care centres located in three different European countries, with a combined total of 2,400 beds, 180,000 patients treated per year, and an annual medicines expenditure of approximately 45 million euros.

The network operates a Central Pharmacovigilance Department (CPD) which, in accordance with EU pharmacovigilance legislation (Regulation (EU) No 1235/2010 as amended by Regulation (EU) No 1027/2012, and Directive 2010/84/EU), is responsible for:

- Detecting and reporting suspected adverse drug reactions (ADRs) to the relevant National Competent Authorities (NCAs).
- Maintaining the Risk Management System for each medicine in use.
- Evaluating emerging safety signals.
- Coordinating with the Pharmacy and Therapeutics Committee at each centre.

The problem. With 2,400 active beds and an average of 8.2 medicines per inpatient, the CPD simultaneously manages up to 19,680 medicine-patient combinations. The pharmacovigilance team comprises twelve members. Signal detection is reactive, relying on voluntary clinician notifications, which have an estimated under-reporting rate of 94% according to EMA data, and on periodic record reviews that take weeks to complete.

The measurable consequence of this problem: over the 2022 to 2024 period, the network identified 234 serious adverse events. A retrospective analysis revealed that 67 of these (28.6%) presented detectable signals in the clinical data between 24 and 72 hours before the event manifested. The current system has no prospective detection capability.

The opportunity. An agentic AI system with continuous monitoring can transform the pharmacovigilance process from reactive to proactive, shifting from detecting events to preventing their consequences. This transformation, however, requires a level of governance that demonstrates, before any clinician acts upon a system recommendation, that the recommendations are reliable, explainable, and auditable.

2.2 System Description

GUARDIAN-Health implements a continuous agentic flow organised across three coordinated modules.

The VIGIL Module runs monitoring agents that connect, via specialised tools, to:

- The electronic prescribing system (simulated using synthetic FHIR data)
- Laboratory result outputs (out-of-range values and significant changes)
- Nursing vital signs records
- Medical progress notes (processed via NLP with semantic embeddings)
- Drug interaction system alerts.

These agents produce **raw signals**: structured observations indicating that something requires attention, without making any clinical judgement as to its significance.

The ASSESS Module receives raw signals via the event bus (Solace) and activates an evaluation crew comprising:

- A **Pharmacologist** agent that assesses the biological plausibility of the causal relationship
- A **Clinician** agent that evaluates the patient's context and alternative diagnoses
- A **Regulatory** agent that determines whether the event is reportable under applicable legislation
- A **Synthesis** agent that produces the final structured evaluation.

Each agent documents its reasoning in a structured format. The final evaluation includes confidence level, supporting evidence, and degree of inter-agent consensus.

The RESPOND Module manages the actions that follow from evaluations:

- Notifications to responsible clinicians, with urgency calibrated to the risk level
- Escalation to human oversight for high-severity events (mandatory HITL)
- Preparation of notification reports to the NCA
- Updates to the knowledge graph in Neo4j
- Complete event and management records in the audit trail.

2.3 Stakeholders and Actors

Actor	Role	Interaction with the System	Required Trust Level
Prescribing physician	Receives alerts, confirms or dismisses signals	Clinical dashboard / notification	Must be able to understand agent reasoning
Clinical pharmacologist	Reviews ASSESS Module evaluations	Pharmacovigilance portal	Requires full evaluation traceability
Medical director	Approves regulatory notifications	Mandatory HITL flow	Digital signature over notification decisions
CPD (Pharmacovigilance)	Manages the system, defines protocols	Administration console	Full control over parameters and thresholds
National Competent Authority	Receives regulatory notifications	Notification API (E2B R3)	Regulatory conformity of format and content
Patient	Subject of the data	No direct interaction	Data protection and non-discrimination
Ethics Committee	System oversight	Governance reports	Access to equity and bias metrics
Data auditors	Compliance verification	Complete audit trail	Traceability of every agentic decision

2.4 Functional and Non-Functional Requirements

Functional Requirements:

- FR-01: The system shall continuously monitor clinical signals for all inpatient subjects, with a maximum latency of 15 minutes from the point of data generation.
- FR-02: The ASSESS Module shall produce structured evaluations including confidence level, supporting evidence, and explicit reasoning for each signal processed.
- FR-03: All events classified as "serious" or "potentially serious" shall undergo mandatory human review before any notification or clinical alert is generated.

- FR-04: The system shall generate draft notifications to the NCA in E2B R3 format for all reportable events, requiring human approval prior to submission.

- FR-05: The audit trail shall record every agent action, every tool invoked, every LLM call with its complete prompt and response, and every HITL decision, with an immutable timestamp and integrity signature.

- FR-06: The system shall maintain an up-to-date knowledge graph in Neo4j containing all detected relationships between medicines, patients, diagnoses, and adverse events.

Non-Functional Requirements:

- NFR-01 (Explainability): Any alert generated shall be accompanied by an explanation in plain language comprehensible to a clinician who is not an AI expert, accessible from the interface in no more than three clicks.

- NFR-02 (Auditability): The audit trail shall be retained for a minimum of ten years, with the capability to reconstruct the complete system state at any point in the past.

- NFR-03 (Availability): The VIGIL Module shall maintain 99.5% availability on a 24/7 basis. The ASSESS and RESPOND modules may operate with graceful degradation.

- NFR-04 (Equity): The system shall be periodically evaluated to detect systematic biases in signal classification across subpopulations defined by age, sex, and ethnicity.

- NFR-05 (Privacy by Design): Patient data shall be pseudonymised in all system contexts except where clinical identification is specifically required, with all access controlled and logged.

2.5 Regulatory and Compliance Constraints

- **GDPR.** Clinical patient data constitutes special category data under Article 9 of the GDPR. Any processing by AI agents requires an explicit legal basis (public interest in healthcare, Article 9(2)(h)), technical pseudonymisation measures, and a Record of Processing Activities (Article 30 GDPR).

- **EU AI Act.** A clinical decision support system with the potential to influence prescribing decisions is classified as a **high-risk AI system** (Annex III, Category 5(a)). This entails a conformity assessment, comprehensive technical documentation, registration in the EU database, a risk management system, and mandatory human oversight throughout the system lifecycle.

- **Pharmacovigilance Legislation.** Regulation (EU) No 1235/2010 as amended, Directive 2010/84/EU, and EMA Good Pharmacovigilance Practices (GVP) Modules. Any system that

generates or facilitates notifications to an NCA becomes part of the national pharmacovigilance system and must comply with applicable reporting timelines (15 days for serious ADRs, 90 days for non-serious) and formats (E2B R3).

- **Medical Device Regulation (EU) 2017/745.** Were the system to be commercialised, it might require CE marking as medical device software (Software as a Medical Device, SaMD) under the Medical Device Regulation, depending on its level of decisional autonomy.

- **WHO Uppsala Monitoring Centre (UMC) Standards.** As a best practice reference applicable across multiple national jurisdictions, the WHO-UMC causality assessment criteria and signal management guidelines inform the evaluation logic of the ASSESS Module.

3. Enterprise Architecture

3.1 Preliminary Phase: Establishing the Framework

The Preliminary Phase of TOGAF 10 establishes the architecture capability before specific architecture work commences. For GUARDIAN-Health, this means defining the following:

Architecture Principles for Agentic Systems in Healthcare:

- P-01 **Integrated Governance.** Every system component, including each agent and each tool, operates within an explicit governance framework. No components exist without an owner, a usage policy, and an oversight mechanism.

- P-02 **Universal Traceability.** No system action, including LLM inference decisions, may occur without leaving an auditable record. Traceability is not a feature; it is a design constraint.

- P-03 **Human Oversight Proportional to Risk.** The level of agent autonomy is inversely proportional to the risk associated with its actions. Agents with the highest risk potential are afforded the least autonomy.

- P-04 **Privacy by Design.** Patient data protection is a first-order design requirement, not a secondary implementation consideration.

- P-05 **Operational Explainability.** The system shall be capable of explaining its decisions in terms comprehensible to each user type: clinician, administrator, regulator, and patient.

- P-06 **Graceful Degradation.** The system shall operate safely under partial failure conditions, prioritising patient safety over agentic system continuity.

Architecture Capability Maturity. For HealthBridge Network, the architecture maturity assessment begins at Level 2 (Defined) for clinical processes but at Level 1 (Initial) for AI systems. The

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GUARDIAN-Health architecture programme includes elevation of AI maturity to Level 3 (Managed) as a production prerequisite.

3.2 Phase A: Architecture Vision

Statement of Architecture Work:

HealthBridge Network requires the transformation of its pharmacovigilance process from reactive to proactive, by implementing an agentic AI system that continuously monitors safety signals, evaluates their clinical significance, and coordinates responses. Human oversight must be maintained for all decisions with the potential for direct patient safety impact. The system must comply with all applicable regulatory requirements, including the EU AI Act, GDPR, and EU pharmacovigilance legislation.

Architecture Vision:

The target state over an 18-month horizon is a digital ecosystem in which:

- 100% of inpatients are under active safety signal monitoring.
- The average signal detection time falls from days (manual periodic review) to minutes (continuous agentic monitoring).
- 100% of regulatory notifications are AI-assisted, with mandatory human oversight.
- A complete, immutable audit trail exists for every system decision.
- The system is documented as a compliant high-risk AI system under the EU AI Act.

Capability Assessment:

Capability	Current State	Target State	Gap
Signal monitoring	Reactive, manual	Proactive, continuous	Critical
ADR evaluation	12 pharmacologists, manual process	AI-assisted with human review	High
Regulatory notification	Semi-structured process	AI-assisted, E2B R3 format	Medium
Decision audit trail	Partial, unstructured	Complete, immutable	High
AI governance	Non-existent	Full framework	Critical
Data interoperability	Proprietary EHRs, no API	FHIR R4, standard API	High

3.3 Phase B: Business Architecture

Business Capability Map:

Business capabilities are organised across four domains.

- **Clinical Domain:** Medicine prescribing and dispensing, clinical monitoring, safety incident management, and care coordination.
- **Pharmacovigilance Domain:** Signal detection, causality assessment, regulatory notification management, authority communication, and prescriber education.
- **AI Governance Domain:** Agent policy definition, decision oversight, model management, bias monitoring, and system audit.
- **Data and Compliance Domain:** Clinical data management, GDPR compliance, access management, regulatory reporting, and evidence retention.

Value Streams Affected:

The primary value stream that GUARDIAN-Health transforms is the chain of Signal Detection, Evaluation, Response, and Learning. In the current model, each step is manual and disconnected. In the target model, agents automate detection and preliminary evaluation, whilst human experts focus on the highest-complexity clinical decisions and regulatory responsibilities.

Roles and Responsibilities (RACI for key decisions):

Decision	VIGIL Agent	ASSESS Agent	Pharmacologist	Medical Director	CPD
Raw signal detection	R	N/A	I	N/A	I
Severity classification	N/A	R	A	N/A	I
HITL activation	N/A	R	N/A	A	I
NCA notification approval	N/A	N/A	R	A	C
Case closure	N/A	N/A	R	I	A

3.4 Phase C: Information Systems Architecture

Application Architecture:

- The GUARDIAN-Health application architecture follows a layered pattern with clear separation of responsibilities.
- **Data Acquisition Layer.** FHIR R4 connectors, HL7 v2 adapters, and Solace Agent Mesh listeners. Transforms data from source systems into the system's internal representation.
- **Agentic Processing Layer.** LangGraph (stateful flow orchestration), CrewAI (evaluation crews), and AutoGen (adversarial evaluation). Implements the logic of the three modules VIGIL, ASSESS, and RESPOND.
- **Data Layer.** Neo4j (clinical knowledge graph), PostgreSQL (transactional data and audit trail), Pinecone/Qdrant (medical literature vector index).
- **Integration Layer.** FastAPI (REST APIs), Solace Agent Mesh (asynchronous messaging), and n8n (automation workflows).
- **Presentation and HITL Layer.** Clinical dashboard (FastAPI and frontend), governance console (system metrics and alerts), and pharmacovigilance portal.
- **Observability Layer.** LangSmith (agentic traces), structured logging, and system metrics.

Application Data Architecture:

Data flows in an event-driven pattern. Clinical systems publish events to Solace; the VIGIL Module consumes and processes them; detected signals are published as new events; the ASSESS Module consumes these and produces evaluations; the RESPOND Module acts on the evaluations. At each step, data is persisted with its complete provenance.

3.5 Phase D: Technology Architecture

Complete Technology Stack:

```
+-------------------------------------------------------------------+
|                      PRESENTATION LAYER                           |
|   FastAPI Clinical Dashboard | Pharmacovigilance Portal | n8n    |
+-------------------------------------------------------------------+
|                  AGENTIC ORCHESTRATION LAYER                      |
|   LangGraph (Agentic Flow) | CrewAI (Crews) | AutoGen (Eval)     |
|               LangChain (LLM + Tool Abstraction)                  |
+-------------------------------------------------------------------+
```

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```
|                      INFERENCE LAYER                              |
|     Groq API (llama-3.3-70b-versatile / llama-3.1-8b-instant)   |
|     HuggingFace (nomic-embed-text-v1, local embeddings)          |
+-------------------------------------------------------------------+
|                     INTEGRATION LAYER                             |
|  Solace Agent Mesh (Events) | FastAPI (REST) | n8n (Workflows)   |
+-------------------------------------------------------------------+
|                       DATA LAYER                                  |
|  Neo4j (Clinical Graph) | PostgreSQL (Transactional + Audit)     |
|  Pinecone (Vector Prod)  | Qdrant (Vector Dev)                    |
+-------------------------------------------------------------------+
|                   OBSERVABILITY LAYER                             |
|  LangSmith (Agentic Traces) | Structured Logging | Metrics       |
+-------------------------------------------------------------------+
```

Infrastructure Decisions:

The entire stack runs in Docker containers (with `docker-compose` for the development environment), deployable to Kubernetes for production. This decision guarantees environment reproducibility, service isolation, and the ability to scale components independently.

LLMs are not executed locally (except for unit testing with small models via optional Ollama). Inference is delegated to Groq, which means that data sent to the LLM must be pseudonymised before leaving the controlled perimeter.

3.6 Architecture Governance and ADRs

Architecture Decision Records (ADRs) are formal records of the rationale behind each significant technical decision. They are governance instruments: they allow future architects to understand the reasoning behind each choice and assess its continued validity as context changes. See Appendix A for the complete ADRs for this project.

4. Data Architecture (DAMA DMBOK2)

4.1 Data Governance: Structure and Responsibilities

DAMA DMBOK2 defines Data Governance as "the exercise of authority, control and shared decision-making (planning, monitoring and enforcement) over the management of data assets." For GUARDIAN-Health, this framework translates into a governance structure specific to clinical data in an agentic AI context.

Data Governance Council (DGC):

The DGC is the supreme governance body for GUARDIAN-Health data, comprising:

- **Data Owner:** Medical Director. Ultimately responsible for the system's clinical data.
- **Clinical Data Steward:** Head of the Central Pharmacovigilance Department. Responsible for data quality and correct use within the clinical context.
- **Technical Data Steward:** Data Architect. Responsible for the technical implementation of data policies.
- **Data Protection Officer (DPO):** Responsible for GDPR compliance.
- **Data Quality Lead:** Responsible for data quality processes.

System Data Domains:

GUARDIAN-Health manages five distinct data domains, each with its own governance policy.

- **Clinical-Patient Domain.** Personal health data. Classification: Highly Confidential. Access: Authorised healthcare personnel with an established care relationship only. Retention: As per applicable national health records legislation. Pseudonymisation is mandatory for use in AI agents.

- **Pharmacological Domain.** Medicine data, prescriptions, and dispensing records. Classification: Confidential. Access: Clinicians, pharmacologists, and the agentic system (without pseudonymisation restrictions). Retention: Ten years.

- **Adverse Events Domain.** Detected signals, evaluations, and regulatory notifications. Classification: Highly Confidential. Access: CPD, responsible clinician, and regulators. Retention: Twenty years (pharmacovigilance regulatory requirement).

- **Agentic Decisions Domain.** System action audit trail, LLM traces, and flow states. Classification: Confidential-Internal. Access: Auditors, system administrators, and the DPO. Retention: Ten years.

- **Medical Knowledge Domain.** Indexed literature, clinical guidelines, and drug information sheets. Classification: Public/Licensed. Access: No restrictions within the system. Retention: Continuous update.

Data Policies:

- DP-01: No identifiable patient data may be sent to an external LLM service. All contexts sent to Groq must use patient pseudonyms (UUID) and must have been reviewed by the data sanitisation function.

- DP-02: Every data field in the system must have an assigned Data Steward, documented in the data catalogue.

- DP-03: Any schema change to the Clinical-Patient Domain or the Adverse Events Domain requires DGC approval before implementation.

- DP-04: Data in the Agentic Decisions Domain is immutable once recorded. Records may not be modified or deleted; they may only be marked as corrected, with the correction record appended.

4.2 Data Architecture and Conceptual Model

Principal Domain Entities:

```
PATIENT --------- has ----------- ADMISSION
    |                                 |
    |                             includes
    |                                 |
    |                          PRESCRIPTION ------- contains ------- MEDICINE
    |                                 |                                   |
    |                             generates                            has
    |                                 |                                   |
    +------- experiences ------- ADVERSE_EVENT                   DRUG_DATASHEET
                                      |
                              evaluated_by
                                      |
                              AGENTIC_EVALUATION
                                      |
                         +-------------+--------------+
                         |                            |
                   RAW_SIGNAL                  NOTIFICATION
                  (VIGIL Module)              (NCA / E2B R3)
```

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Graph Model in Neo4j:

```
-- Principal nodes

(:Patient {pseudo_id, age_band, sex, ethnicity_group})

(:Medicine {active_substance, brand_name, atc_code, route_of_administration})

(:Prescription {id, start_date, end_date, dose, frequency})

(:Diagnosis {icd10_code, description})

(:AdverseEvent {id, detection_date, type, severity, status})

(:AgenticEvaluation {id, date, confidence_level, agent_consensus})

(:Notification {id, date, type, status, nca_case_number})

-- Relationships

(:Patient)-[:RECEIVES]->(:Prescription)

(:Prescription)-[:CONTAINS]->(:Medicine)

(:Patient)-[:HAS_DIAGNOSIS]->(:Diagnosis)

(:Patient)-[:EXPERIENCES]->(:AdverseEvent)

(:AdverseEvent)-[:POSSIBLY_CAUSED_BY]->(:Medicine)

(:AdverseEvent)-[:EVALUATED_BY]->(:AgenticEvaluation)

(:AgenticEvaluation)-[:GENERATES]->(:Notification)

(:Medicine)-[:INTERACTS_WITH]->(:Medicine)

(:Medicine)-[:CONTRAINDICATED_IN]->(:Diagnosis)
```

Relational Model in PostgreSQL:

The principal tables of the relational schema cover: `patients` (pseudonymised master data), `admissions`, `prescriptions`, `laboratory_results`, `raw_signals`, `evaluations`, `notifications`, `audit_log` (immutable, append-only), `agent_decisions` (each agentic decision with trace), `hitl_reviews` (human reviews with decision and justification), and `langgraph_checkpoints` (persistent agent graph state).

4.3 Master Data and Reference Data Management

The system's master data includes the medicine catalogue (with ATC and national codes), the diagnosis table (ICD-10), the healthcare personnel catalogue (for responsibility assignment), and the hospital centres in the network.

These data are managed with a simplified Master Data Management (MDM) process: a single source of truth per master entity, a documented synchronisation process with source systems, and record versioning to allow historical reconstruction.

The reference data critical to agent operation includes pharmacovigilance clinical guidelines, drug interaction databases (simulated with synthetic data), and the EMA severity classification criteria. These data are loaded into Pinecone/Qdrant for semantic access by agents.

4.4 Data Quality in Agentic Contexts

Data quality in an agentic system carries an additional dimension compared to traditional systems: the quality of input data directly affects the quality of agent reasoning. An agent reasoning over incorrect data may produce plausible but erroneous evaluations, and the nature of the system makes these errors difficult to detect without active monitoring.

Quality Dimensions Managed:

- DQ-01 **Completeness.** Data sent to agents must meet a minimum completeness level. The system verifies, before activating the ASSESS Module, that the patient, prescription, and event data contain the fields required for evaluation.

- DQ-02 **Timeliness.** In pharmacovigilance, a correct but late piece of data may be worthless. The system monitors the latency of each data source and reduces the confidence level of evaluations based on data whose latency exceeds defined thresholds.

- DQ-03 **Consistency.** Data relating to the same patient arriving from different systems (EHR, pharmacy, laboratory) must be reconciled. The system maintains a patient identity reconciliation process using the Neo4j graph.

- DQ-04 **Contextual Accuracy.** This dimension is specific to agentic systems. It measures whether the data in the agent's context is representative of the actual clinical situation. It is monitored through the agreement rate between agentic evaluations and subsequent human reviews.

Data Quality Framework for Agent Contexts:

Before including any data in an agent's context (the prompt), the system executes a validation process that checks for required field completeness, consistency with the patient's historical data, absence of directly identifying data (pseudonymisation), and data freshness relative to the maximum latency policy.

4.5 Lineage, Metadata, and Traceability

Data lineage in GUARDIAN-Health is bidirectional: for every signal detected by the system, one can trace backwards to the source data that generated it. And for every source datum, one can trace forwards to the agentic decisions in which it played a role.

Minimum Required Metadata:

Every datum in the system carries a minimum set of metadata: `source_system` (the system that generated the datum), `ingestion_timestamp` (the moment of ingestion into the system), `data_as_of` (the moment to which the datum refers, which may differ from the ingestion timestamp), `quality_score` (the quality score calculated at ingestion), `steward` (the responsible Data Steward), `classification` (confidentiality classification), and `used_in_agent_contexts` (the list of agentic contexts where this datum was included, for auditability).

Data Catalogue:

The data catalogue is implemented in PostgreSQL and documents every data entity in the system with its business definition, owner, classification, technical schema, and lineage. The catalogue is the authoritative source of truth for auditors and the DPO.

4.6 Data Pipeline: Medallion Architecture in Healthcare

The data pipeline follows the Medallion architecture (Bronze, Silver, Gold) adapted to the clinical context.

Bronze Layer (Raw Data). Data exactly as received from source systems, without transformation. This includes complete HL7 v2 messages, raw FHIR responses, and laboratory system exports in proprietary format. This layer is retained for forensic audit and reproducibility. Data from this layer is never used directly in agents.

Silver Layer (Cleansed and Conformed Data). Data normalised to the system's conceptual model, quality-validated, pseudonymised, and enriched with cross-references (ATC codes, canonical ICD-10 codes). This layer feeds Neo4j and the PostgreSQL relational tables. Monitoring agents (VIGIL Module) consume data from this layer.

Gold Layer (Agentic Data). Data prepared specifically for agent consumption: pre-built contexts, clinical history summaries, and structured signals. Also includes the vector embeddings of indexed medical documentation in Pinecone/Qdrant. Evaluation agents (ASSESS Module) consume data from this layer.

4.7 Privacy, Security, and Data Compliance

Pseudonymisation:

The system employs a two-layer pseudonymisation scheme. The outer layer assigns a permanent UUID to each patient, stored in a mapping table with very restricted access. The inner layer generates temporary session identifiers for each agent context, which expire and are rotated. Data sent to Groq uses only the session identifier, which cannot be re-linked to the patient without accessing the pseudonymisation system.

Access Control:

The system implements role-based access control (RBAC) with field-level granularity for patient data. Agents have specific access roles permitting only the data necessary for their function (principle of least privilege).

Retention and Deletion:

Retention policies are implemented as automated processes with human oversight. The GDPR right to erasure is implemented via enhanced pseudonymisation (destruction of the mapping key) rather than physical deletion, in order to preserve the statistical utility of historical data.

5. Agentic AI Architecture

5.1 Agent Patterns and Selection Rationale

GUARDIAN-Health employs four distinct agentic patterns, each selected for its suitability to governance requirements and the specific characteristics of its module.

ReAct (Reasoning and Acting). This is the base pattern for VIGIL Module agents. The agent alternates between reasoning (interpreting available data) and acting (invoking tools to obtain additional data). This pattern is well suited to the detection phase because it is transparent: the agent's reasoning is explicit at each step, which facilitates auditability.

Plan-and-Execute. Employed in the ASSESS Module for complex signal evaluation. The agent first produces an explicit evaluation plan (what information it needs, what criteria it will apply, what simulated experts it will consult), then executes each step of the plan. This pattern improves the coherence of complex evaluations and produces a reasoning artefact (the plan) that can be reviewed by a human.

Multi-Agent Debate. Implemented via AutoGen in the ASSESS Module for ambiguous classification cases. Two agents adopt opposing positions (a proposing agent favouring serious classification, and a sceptical agent questioning causality) and debate in a structured format. The debate process and the final synthesis are recorded as evidence of the reasoning.

Supervisor-Worker. Employed in the RESPOND Module with LangGraph. A Supervisor agent coordinates specialised workers (notification drafter, clinician communication agent, graph update agent). The Supervisor maintains global state and makes routing decisions.

5.2 Multi-Agent Topology: Roles and Responsibilities

```
+-------------------------------------------------------------------+
|                   VIGIL MODULE (LangGraph)                        |
|                                                                   |
|  +-------------------+    +-------------------+                  |
|  | Monitoring Agent  |    | Monitoring Agent  |                  |
|  | Medicines         |    | Laboratory        |                  |
|  +--------+----------+    +--------+----------+                  |
|           +-------------------+----+                              |
|                       +-------v-------+                           |
|                       | Correlation   |                           |
|                       | Agent         | --publishes raw signals-> |
|                       +---------------+                           |
+-------------------------------------------------------------------+
                            | Solace Agent Mesh
                            v
+-------------------------------------------------------------------+
|                ASSESS MODULE (CrewAI + AutoGen)                   |
|                                                                   |
|  +-------------+  +--------------+  +---------------------+     |
|  | Pharmacolo- |  | Clinician    |  | Regulatory          |     |
|  | gist Agent  |  | Agent        |  | Agent               |     |
|  +------+------+  +------+-------+  +---------+-----------+     |
|         +------------------+-----------------+                   |
|                      +------v------+                              |
|                      | Synthesis   |                              |
|                      | Agent       | -> Structured evaluation     |
|                      +------+------+                              |
|                             |                                     |
|                      High severity? -> Mandatory HITL             |
+-------------------------------------------------------------------+
                            | (post-HITL if applicable)
                            v
+-------------------------------------------------------------------+
|                RESPOND MODULE (LangGraph)                         |
|                                                                   |
|  +----------------+                                               |
|  | Supervisor     +-> Clinician Notification Agent               |
|  | Agent          +-> NCA Draft Agent (E2B R3)                   |
|  |                +-> Neo4j Update Agent                         |
|  +----------------+                                               |
+-------------------------------------------------------------------+
```

5.3 Orchestration with LangGraph

LangGraph implements the VIGIL Module and the RESPOND Module as state graphs. The design of the state graph is the most important governance decision in the agentic architecture.

Graph State (TypedDict):

```
class GuardianState(TypedDict):
    # Identity and traceability
    session_id: str                     # UUID of the agentic session
    patient_pseudo_id: str              # Pseudonymised patient ID
    trace_id: str                       # LangSmith trace ID

    # Clinical context data
    clinical_context: dict              # Sanitised patient data
    prescriptions_active: list          # Active prescriptions (pseudonymised)
    lab_results_recent: list            # Recent laboratory results

    # Signals and evaluations
    raw_signals: list[RawSignal]        # Detected raw signals
    evaluation: Optional[Evaluation]    # ASSESS Module evaluation

    # Flow control and governance
    requires_hitl: bool                 # Human review required?
    hitl_decision: Optional[str]        # Human decision if applicable
    hitl_reviewer_id: Optional[str]     # Reviewing human's ID
    hitl_timestamp: Optional[str]       # Review timestamp

    # Audit trail
    agent_actions: list[AgentAction]    # Record of every agent action
    tool_calls: list[ToolCall]          # Record of every tool invoked
    llm_calls: list[LLMCall]            # Record of every LLM call

    # Flow state
    current_module: str                 # Active module
    flow_status: str                    # Flow status
    error_state: Optional[str]          # Error if present
```

The state includes the audit trail explicitly as part of the graph state. This is not secondary logging: it is the governance mechanism integrated directly into the agentic flow.

VIGIL Graph Nodes:

```
vigil_graph.add_node("initialise", initialise_node)
vigil_graph.add_node("fetch_clinical_context", fetch_context_node)
vigil_graph.add_node("sanitise_data", sanitise_data_node)
```

```
vigil_graph.add_node("monitor_prescriptions", monitor_prescriptions_node)
vigil_graph.add_node("monitor_lab_results", monitor_lab_node)
vigil_graph.add_node("correlate_signals", correlate_signals_node)
vigil_graph.add_node("publish_signals", publish_signals_node)
vigil_graph.add_node("update_audit_trail", update_audit_node)

# Conditional edges for governance
vigil_graph.add_conditional_edges(
    "correlate_signals",
    route_by_signal_quality,  # Verifies minimum quality before publishing
    {
        "publish": "publish_signals",
        "low_quality_halt": "update_audit_trail"  # Documents the halt
    }
)
```

Checkpointing for Governance:

The LangGraph checkpointer persists the complete state to PostgreSQL after each node. This allows: resumption of an interrupted flow from the last checkpoint; audit of the exact system state at any point in the process; full flow replay for forensic investigation; and HITL implementation as a graph interrupt (the graph pauses, awaits the human decision, and resumes).

5.4 Crews with CrewAI and AutoGen

Evaluation Crew (CrewAI):

```
assessment_crew = Crew(
    agents=[
        pharmacologist_agent,   # Assesses biological plausibility
        clinician_agent,        # Assesses patient context
        regulatory_agent,       # Assesses reportability
        synthesis_agent         # Produces the final evaluation
    ],
    tasks=[
        causality_assessment_task,
        severity_classification_task,
        notifiability_determination_task,
        final_evaluation_task
    ],
    process=Process.sequential,
    verbose=True,
    output_log_file="crew_assessment_log.json"  # Governance: crew log
)
```

Each crew task produces a Pydantic-typed output. The output model for `final_evaluation_task` includes the confidence level, the synthesis of each agent's reasoning, inter-agent discrepancies (where present), and the final recommendation with its justification.

Adversarial Debate (AutoGen):

For signals with discordant evaluations across crew agents, an AutoGen debate process is activated. One agent adopts the role of "defender of the proposed classification" whilst another adopts the role of "critical evaluator." The debate follows a structured protocol with a maximum number of rounds and defined termination criteria. The complete debate transcript is stored as evidence of the reasoning.

5.5 Memory Design: Short-Term, Long-Term, and Episodic

- **Short-Term Memory.** The LangGraph graph state is the system's short-term memory. It contains the active context of the current agentic session.

- **Long-Term Memory.** Neo4j is the system's relational long-term memory. It contains the accumulated clinical knowledge graph: relationships between medicines, patients, diagnoses, and adverse events. Agents can query the graph to find similar patterns in the hospital network's history.

- **Episodic Memory.** Pinecone/Qdrant stores embeddings of previous evaluations, clinical guidelines, and relevant medical literature. Agents use semantic search to retrieve similar episodes (previous evaluations of comparable signals) that enrich their reasoning.

- **Agent Working Memory.** For each LLM invocation, the context is constructed explicitly by combining elements from the three memory types above, following a context construction template that guarantees: the context does not exceed model limits; data is correctly pseudonymised; and provenance metadata is included.

5.6 Human-in-the-Loop (HITL): Design and Rationale

HITL is not a concession to distrust in agents: it is a first-order architecture component, defined by regulatory requirements and governance principles.

HITL Decision Matrix:

Action Type	Agent Autonomy	HITL Level	Rationale
Raw signal detection	Full	None	Low risk, high frequency, reviewable
Classification as mild	Full	Informational notification	Reversible impact
Classification as moderate	Proposal	24-hour confirmation	Requires clinical validation
Classification as serious	Draft	Prior approval	High impact, not reversible
NCA notification	Draft	Mandatory approval and signature	Regulatory act
Direct patient communication	Prohibited	N/A	Outside system scope

HITL Implementation in LangGraph:

```
def check_hitl_requirement(state: GuardianState) -> str:
    evaluation = state["evaluation"]
    if evaluation.severity in ["SERIOUS", "POTENTIALLY_SERIOUS"]:
        state["requires_hitl"] = True
        return "hitl_required"  # Graph is interrupted here
    return "proceed_to_respond"

# The interrupt is implemented with interrupt_before
guardian_graph = StateGraph(GuardianState)
guardian_graph.add_node("await_hitl_decision", await_hitl_node)

# LangGraph pauses the graph at this node until `resume` is called
# with the human decision as input
```

The human decision is recorded in the graph state with: reviewer ID, timestamp, decision (approve / reject / modify), and free-text justification. This record is part of the permanent audit trail.

5.7 Tools and External APIs

System agents have access to a specifically designed and approved set of tools. They do not have access to generic internet search tools or arbitrary code execution.

Clinical Data Tools:

- `get_patient_context(pseudo_id)`: Retrieves the pseudonymised clinical context from the Silver Layer.
- `get_active_prescriptions(pseudo_id)`: Returns active prescriptions with ATC coding.
- `get_lab_results(pseudo_id, days=7)`: Recent laboratory results with reference ranges.
- `query_clinical_graph(cypher_query, params)`: Executes Cypher queries on Neo4j (with an allowed-operations whitelist).

Medical Knowledge Tools:

- `search_pharmacovigilance_guidelines(query)`: Semantic search over EMA/NCA guidelines.
- `search_similar_cases(signal_description)`: Retrieves similar previous evaluations from Pinecone/Qdrant.
- `get_drug_interactions(atc_codes)`: Queries the interaction database (local, not external).

Flow Management Tools:

- `classify_severity(event_description, criteria)`: Applies standard classification criteria.
- `generate_e2b_draft(evaluation)`: Generates an E2B R3 notification draft.
- `notify_clinician(clinician_id, urgency, message)`: Sends a notification (with audit log entry).
- `update_knowledge_graph(entities, relations)`: Updates Neo4j with detected entities.

All tools include automatic logging: every invocation records the input parameters, the output, execution time, and the invoking agent.

5.8 Event-Driven Messaging: Solace Agent Mesh

Solace Agent Mesh serves as the communication fabric between the three modules. Messaging topics follow a structured hierarchy:

```
guardian/v1/signals/raw/{centre_id}/{patient_pseudo_id}
guardian/v1/signals/evaluated/{severity}/{patient_pseudo_id}
guardian/v1/hitl/required/{case_id}
guardian/v1/hitl/decision/{case_id}
guardian/v1/notifications/clinician/{clinician_id}
guardian/v1/notifications/nca/{case_id}
guardian/v1/audit/{module}/{event_type}
```

The topic hierarchy carries governance meaning. The `guardian/v1/audit/` topics are write-only for modules and read-only for the audit system. The `guardian/v1/hitl/decision` topic requires specific human role authentication.

5.9 Agentic Observability: LangSmith and Agent Traces

LangSmith automatically captures traces for all LangChain chains and LangGraph graphs. For GUARDIAN-Health, LangSmith configuration is part of the governance framework, not merely a debugging aid.

Trace Metadata:

Every trace is enriched with governance metadata:

```
with trace(
    project_name="guardian-health",
    tags=["module:vigil", "centre:north"],
    metadata={
        "patient_pseudo_id": state["patient_pseudo_id"],
        "session_id": state["session_id"],
        "data_classification": "confidential",
        "hitl_required": state["requires_hitl"],
        "regulatory_case_id": state.get("regulatory_case_id")
    }
):

    # Agent execution
```

These traces are queryable and constitute part of the regulatory audit trail.

6. Governance and Accountability by Design

6.1 Why Governance Must Begin on Day Zero

This is the central thesis of GUARDIAN-Health, and it is worth articulating with technical precision, because the industry tends to treat it as a management problem rather than what it actually is: an architecture problem.

Retroactive governance does not scale.

When governance is added to an already-built system, it is added as a layer of external instrumentation: application logs, monitoring dashboards, periodic output reviews. This external instrumentation has a fundamental problem: it cannot capture the internal reasoning of the system. It can tell you what the agent decided, but not why it reasoned in that way, with what data, what alternatives it considered, and what factors determined its choice. In agentic AI systems, the "why" is precisely what auditors, regulators, and clinicians need to understand.

Governance integrated from the design phase captures the reasoning at the moment it occurs, as part of the system state, not as a secondary artefact. This is what the LangGraph state design does in GUARDIAN-Health: the audit trail is not a separate system observing the agent; it is part of the agent's state.

The trust deficit accumulates exponentially.

An agentic system operating in a high-risk domain without visible governance generates a trust deficit between users (clinicians, regulators) and the system that grows with each unexplained decision. Recovering that trust requires demonstrating not only that the system functions correctly now, but that it has functioned correctly from the outset. Without an audit trail from the beginning, that demonstration is impossible.

Governance debt has a point of no return.

In traditional software systems, technical debt can be repaid through refactoring. In agentic AI systems with a decisional history, governance debt cannot be repaid retrospectively: the traces of decisions that were not recorded cannot be recreated, the lineage of data that was not captured cannot be reconstructed. The cost of governance from day zero is the investment in design. The cost of retroactive governance, where it is achievable at all, is typically greater than the cost of having designed the system correctly from the outset. In many cases, retroactive governance is simply not achievable.

The EU AI Act makes governance a legal constraint, not a recommended practice.

For high-risk AI systems (the category into which GUARDIAN-Health would fall), the EU AI Act requires comprehensive technical documentation, a risk management system, traceability logging (Article 12), human oversight (Article 14), and a conformity assessment before deployment. These obligations can only be verified if the system was designed to produce the required evidence. One cannot build the system first and then attempt to demonstrate conformity if the system was not designed for that purpose.

6.2 Risks of Absent Agentic Governance in Early Stages

Risk 1: Cumulative Decisional Opacity

Without traceability structures designed from the outset, agent decisions accumulate as "black boxes": the system produces outputs, but the reasoning chain that led to each output is unavailable for review. In pharmacovigilance, where an incorrect classification can have clinical consequences, this opacity is unacceptable from the very first day of operation.

Risk 2: Undetected Behavioural Drift

Agentic systems can experience gradual and imperceptible drift in their behaviour over time: model drift, prompt drift, and context drift. Without integrated governance monitoring, this drift is not detected until its consequences become observable in the form of an increasing rate of incorrect evaluations, systematic classification bias, or increased processing times. With integrated governance, drift indicators are visible in real time.

Risk 3: Indeterminate Legal Liability

When an agentic AI system contributes to an undetected adverse event or an incorrect regulatory notification, the central legal question is: who is responsible, and to what degree did the system contribute to that outcome? Without an audit trail of agent decisions, the data processed, and the human oversight performed, this question has no verifiable answer. Liability is left indeterminate, exposing all actors (the hospital, the software developer, the clinicians who acted on the alerts) to litigation without clear resolution.

Risk 4: Impossibility of Regulatory Compliance

The EU AI Act, the GDPR, and pharmacovigilance legislation each require forms of documentation that cannot be generated retrospectively. If the system does not record, from the outset, the data processed, the models used, the decisions taken, and the oversight performed, it cannot demonstrate compliance when a regulator requires it, which may occur at any time.

Risk 5: Erosion of Clinical Trust Capital

Clinicians are demanding and sceptical users. If an AI system produces alerts that cannot be explained in a comprehensible manner, clinicians learn to ignore them. The alert dismissal rate in clinical decision systems without adequate explainability can exceed 90% within weeks of deployment. A pharmacovigilance system that its users ignore is worse than having no system at all.

Risk 6: Uncontrolled Bias Amplification

Large language models inherit biases from their training data. An agent evaluating the severity of adverse events may carry systematic biases related to patient characteristics. Without integrated equity monitoring from the outset, these biases are amplified with each operational cycle: the system acts on its biased evaluations, those actions generate data, and that data reinforces the bias in subsequent cycles.

6.3 A Governance Framework for Agentic Systems

GUARDIAN-Health implements a four-layer governance framework covering the full arc from design to ongoing operation.

- **Layer 1: Design Governance.** Governance policies are defined as architecture constraints before a single line of code is written. These include the HITL matrix (which decisions require human oversight and of what type), data policies (which data agents may access and under what conditions), tool constraints (which tools agents are permitted to use), and state design principles (what information must reside in the graph state to guarantee auditability).

- **Layer 2: Data Governance.** Implements the DAMA DMBOK2 framework detailed in Chapter 4, covering data classification, access policies, data quality, lineage, and retention and secure disposal processes.

- **Layer 3: Operational Governance.** Continuous monitoring of the system's governance metrics: agent-human agreement rate on HITL evaluations, rate of evaluations with low confidence (signal of degraded quality), distribution of classifications by subpopulation (signal of bias), latency per module, and tool error rate.

- **Layer 4: Compliance Governance.** Periodic regulatory compliance verification processes: audit trail audits, reviews of technical documentation required by the EU AI Act, activity reports to the DPO, and Ethics Committee reviews.

6.4 Traceability of Agentic Decisions: The Audit Trail

The GUARDIAN-Health audit trail is an append-only database in PostgreSQL with the following integrity guarantees:

- **Immutability.** Records are cryptographically signed at the moment of insertion. Any subsequent modification is detectable.
- **Completeness.** Each LangGraph node automatically generates an audit trail entry before execution (pre-action) and after completion (post-action). No system action exists without a record.
- **Full context.** Each audit trail entry includes the complete agent state at that moment, not merely a summary. This allows exact reconstruction of any agentic session.
- **Chain linkage.** Each entry references the previous entry via its hash, creating a chain of evidence that guarantees no intermediate entries have been removed.
- **Audit Trail Schema:**

```
CREATE TABLE audit_trail (
    id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    sequence_num    BIGINT NOT NULL,  -- Global, immutable sequence
    session_id      UUID NOT NULL,
    recorded_at     TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    module          VARCHAR(20) NOT NULL,  -- VIGIL, ASSESS, RESPOND
    agent_id        VARCHAR(100) NOT NULL,
     action_type        VARCHAR(50)  NOT  NULL,   --  LLM_CALL,  TOOL_USE,
HITL_REQUEST, etc.
   action_detail   JSONB NOT NULL,        -- Complete action parameters
    state_snapshot  JSONB NOT NULL,        -- Complete agent state
    result          JSONB,                 -- Action result
     previous_hash    VARCHAR(64) NOT NULL,  -- SHA-256 hash of previous
record
   record_hash     VARCHAR(64) NOT NULL,  -- SHA-256 hash of this record
    CONSTRAINT audit_trail_sequence UNIQUE (sequence_num)
);

--  The  table  is  insert-only;  DELETE  and  UPDATE  are  revoked  from  the
agent role

REVOKE DELETE, UPDATE ON audit_trail FROM guardian_agent_role;
```

6.5 Explainability and Transparency

Every evaluation produced by the ASSESS Module includes a natural language explanation generated by the Synthesis Agent. This explanation follows a structured template covering: what was observed (relevant clinical data), what this suggests (the Pharmacologist Agent's interpretation), what alternatives were considered (the Clinician Agent's interpretation), what the regulatory framework requires (the Regulatory Agent's assessment), and what the recommendation is and at what confidence level.

The explanation is presented in the interface in two versions: a technical version for pharmacologists (with full reasoning detail) and a summary version for prescribing physicians (focused on the recommended action and its essential clinical justification).

7. Implementation Plan

7.1 Development Environment and Containers

Following the principle of not installing software directly on the host operating system, the development environment is managed entirely via Docker containers and Python virtual environments.

Container Structure (docker-compose):

```
services:
  neo4j:
    image: neo4j:5.x-enterprise
    environment:
      NEO4J_PLUGINS: '["apoc"]'
    ports: ["7474:7474", "7687:7687"]

  postgres:
    image: postgres:16
    volumes: ["postgres_data:/var/lib/postgresql/data"]
    ports: ["5432:5432"]

  qdrant:
    image: qdrant/qdrant:latest
    ports: ["6333:6333"]

  n8n:
    image: n8nio/n8n:latest
    ports: ["5678:5678"]
    environment:
      N8N_BASIC_AUTH_ACTIVE: "true"
```

```
  solace:
    image: solace/solace-pubsub-standard:latest
    ports: ["8080:8080", "55555:55555", "1883:1883"]

  guardian-api:
    build: ./api
    ports: ["8000:8000"]
    depends_on: [postgres, neo4j, qdrant]
```

Python Virtual Environment:

A new `.venv-guardian` virtual environment is created specifically for this case study, following the pattern established in the bootcamp. The environment includes: `langchain`, `langgraph`, `crewai`, `pyautogen`, `langchain-groq`, `langchain-neo4j`, `langchain-qdrant`, `langchain-pinecone`, `langchain-huggingface`, `fastapi`, `uvicorn`, `solace-pubsub-plus`, `pydantic>=2.0`, `sqlalchemy`, and `psycopg2-binary`.

7.2 Repository Structure

```
guardian-health/
├── langgraph/
├── crewai/
├── README.md
├── ARCHITECTURE.md
├── docker-compose.yml
├── .env.example
├── requirements.txt
├── docs/
│   ├── adrs/
│   └── diagrams/
├── data/
│   ├── synthetic/      (synthetic FHIR patient data)
│   ├── seed/           (initial data loading scripts)
│   └── knowledge/      (clinical guidelines and drug information)
├── src/
│   ├── guardian/
│   │   ├── __init__.py
│   │   ├── config.py
│   │   ├── state.py
│   │   ├── models.py
│   │   └── governance/
│   │       ├── audit.py
│   │       ├── hitl.py
│   │       ├── sanitiser.py
│   │       └── tracer.py
```

```
│   ├── modules/
│   │   ├── vigil/
│   │   ├── assess/
│   │   └── respond/
│   ├── tools/
│   ├── data/
│   │   ├── neo4j/
│   │   ├── postgres/
│   │   └── vector/
│   └── api/
└── tests/
    ├── unit/
    ├── integration/
    └── governance/
```

7.3 Phased Development Roadmap

- **Phase 0: Governance Foundations (Week 1).** Before a single agent is written, the governance infrastructure is implemented: the audit trail schema in PostgreSQL, the pseudonymisation system, LangSmith configuration, and the base LangGraph state with its traceability structure. This sequencing is deliberate. Governance first; agents second.

- **Phase 1: Data Infrastructure (Weeks 1 to 2).** Synthetic data loading, Neo4j configuration with the clinical schema, document indexing in Pinecone/Qdrant, and the data quality validation pipeline.

- **Phase 2: VIGIL Module (Weeks 2 to 3).** Implementation of the LangGraph monitoring graph with its clinical data access tools and the signal publication system via Solace.

- **Phase 3: ASSESS Module (Weeks 3 to 4).** Implementation of the CrewAI evaluation crew, the AutoGen debate process for ambiguous cases, and the HITL mechanisms.

- **Phase 4: RESPOND Module and API (Weeks 4 to 5).** Implementation of the response module, the FastAPI, the governance dashboard, and n8n automation workflows.

- **Phase 5: Integration and Validation (Weeks 5 to 6).** Full integration testing, audit trail validation against audit scenarios, and generation of conformity documentation.

8. Appendices

A. Architecture Decision Records (ADRs)

- **ADR-001: LangGraph as Agentic Flow Orchestrator** Status: Accepted. Context: The system requires stateful agentic flow with persistent state and the capability for HITL as a graph interrupt. Decision: LangGraph over alternatives such as Prefect or Airflow, because it provides native support for stateful LLM agents, HITL interruptions, and PostgreSQL checkpointing. Consequences: LangGraph learning curve. Dependency on the LangChain ecosystem. Benefit: integrated traceability and HITL as a first-class citizen of the flow.

- **ADR-002: PostgreSQL for Audit Trail over Time-Series Databases** Status: Accepted. Context: The audit trail requires immutability, referential integrity with clinical data, and the capability for complex audit queries. Decision: PostgreSQL with an append-only table over InfluxDB or Cassandra, because audit queries require JOINs with clinical data that are more natural in SQL, and PostgreSQL supports the REVOKE constraint to enforce append-only at the database level. Consequences: PostgreSQL may not scale as well as time-series databases at extreme volume. For the current scale (three-centre network), it is appropriate.

- **ADR-003: Pseudonymisation Rather Than Full Anonymisation** Status: Accepted. Context: The GDPR permits use of health data under a legitimate legal basis but requires appropriate technical measures. Full anonymisation would destroy the system's utility for individual pharmacovigilance. Decision: Two-layer pseudonymisation (permanent UUID plus temporary session ID) rather than full anonymisation, because full anonymisation is incompatible with the function of detecting and managing adverse events in individual patients. Consequences: Pseudonymised data remains personal data under the GDPR. Requires additional controls (mapping table with very restricted access, access auditing).

- **ADR-004: Groq as Primary LLM Provider** Status: Accepted. Context: The system requires low-latency LLM inference for near-real-time processing. Local models are too slow for the available hardware. Decision: Groq (llama-3.3-70b-versatile) over OpenAI GPT-4, because Groq's latency is significantly lower, the cost is lower, and model sovereignty (Llama is open-weights) is preferable in a regulated context. Consequences: Data sent to Groq leaves the controlled perimeter. Mitigation: strict pseudonymisation policy before any data is sent to Groq (DP-01).

- **ADR-005: Solace Agent Mesh for Inter-Module Messaging** Status: Accepted. Context: The three system modules are semi-independent and communicate via asynchronous events. Decision: Solace Agent Mesh over Kafka or RabbitMQ, because Solace is designed specifically for agentic AI architectures with support for agentic messaging patterns, and is available as a Docker image for development. Consequences: Solace has a steeper learning

curve than Kafka. The maturity of the Solace Agent Mesh ecosystem for agentic AI is lower than that of Kafka. Revisable for production.

B. Glossary

- **Agent.** A software entity that uses an LLM to reason and make decisions, with access to tools for acting on its environment.
- **Audit Trail.** A chronological, immutable, and complete record of all system actions and decisions, designed to support internal and external audits.
- **Crew (CrewAI).** A group of specialised agents that collaborate in solving a complex task, with defined roles and a coordination process.
- **Data Lineage.** The traceability of a datum's origin, transformations, and usage throughout its lifecycle.
- **Data Steward.** A person responsible for the quality, correct use, and policy compliance of a specific data domain.
- **E2B R3.** The international standard (ICH E2B) for the electronic transmission of individual case safety report (ICSR) information in pharmacovigilance.
- **Adverse Drug Reaction (ADR).** An unwanted and harmful response to a medicine.
- **FHIR R4.** Fast Healthcare Interoperability Resources. The international standard for electronic health data exchange.
- **Graph RAG.** Retrieval-Augmented Generation over a graph database, where retrieval exploits graph relationships.
- **HITL (Human-in-the-Loop).** A design pattern in which a human participates in the decision cycle of an automated system, particularly for high-risk decisions.
- **LangGraph.** A LangChain framework for building stateful agentic flows as directed graphs with checkpointing and HITL support.
- **Medallion Architecture.** A data pipeline organisation pattern in Bronze (raw), Silver (cleansed), and Gold (consumption-ready) layers.
- **NCA (National Competent Authority).** The national regulatory authority responsible for pharmacovigilance in a given country.

- **ReAct (Reasoning and Acting).** An agent pattern that alternates between reasoning (thinking about the situation) and acting (invoking tools).

- **Solace Agent Mesh.** An event-driven messaging infrastructure designed for agentic AI architectures.

- **TOGAF 10.** The Open Group Architecture Framework, version 10. An enterprise architecture framework.

- **DAMA DMBOK2.** Data Management Body of Knowledge, second edition. The reference guide for data management and governance.

C. References

Frameworks and Standards:

- The Open Group: TOGAF Standard, Version 10 (2022)
- DAMA International: DAMA-DMBOK: Data Management Body of Knowledge, 2nd Edition
- ICH: Guideline E2B(R3), Electronic Transmission of Individual Case Safety Reports (2013)
- European Medicines Agency: Good Pharmacovigilance Practices (GVP) Modules
- World Health Organisation: WHO Programme for International Drug Monitoring
- EMA Guideline on Good Pharmacovigilance Practices, Module IX: Signal Management.

Regulation:

- Regulation (EU) 2024/1689 (EU AI Act)
- Regulation (EU) 2016/679 (GDPR)
- Regulation (EU) No 1235/2010 as amended by Regulation (EU) No 1027/2012
- Directive 2010/84/EU (Pharmacovigilance)
- Regulation (EU) 2017/745 (Medical Device Regulation).

Technologies:

- LangGraph Documentation: https://langchain-ai.github.io/langgraph/
- CrewAI Documentation: https://docs.crewai.com/
- Microsoft AutoGen: https://microsoft.github.io/autogen/
- Neo4j Graph Data Science: https://neo4j.com/docs/graph-data-science/
- Solace Agent Mesh: https://solace.com/solutions/artificial-intelligence/
- Groq Documentation: https://console.groq.com/docs/
- EMA Pharmacovigilance Portal: https://www.ema.europa.eu/en/human-regulatory/post-authorisation/pharmacovigilance